Modeling Vibrations by Bessel Functions

1. Vibrations
   - a mathematical model
2. The Wave Equation
   - striking a circular drum
   - separation of variables
3. Bessel Functions
   - solving the wave equation in polar coordinates
   - imposing the conditions
   - an orthogonal basis

MCS 472 Lecture 40
Industrial Math & Computation
Jan Verschelde, 22 April 2026

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modeling vibrations

To model vibrations, use *the wave equation* $\displaystyle\frac{\partial^2 u}{\partial t^2} = \nabla^2 u$.

For example, to model a vibrating string

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad \text{for} \quad x \in [0,1], \quad t \geq 0,$$

subject to the initial conditions:

1. $u(x,0) = f(x)$, profile of a string,
2. $u_t(x,0) = 0$, no velocity,

and boundary conditions:

3. $u(0,t) = 0$, string is fixed at the left end,
4. $u(1,t) = 0$, string is fixed at the right end.

The method of separation of variables provides only a weak solution.

traveling waves

Exercise 1:
Use the method of characteristic lines to show that

$$u(x,t) = f(x - ct) + g(x + ct)$$

satisfies the one dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad \text{for} \quad x \in [0,1], \quad t \geq 0,$$

where c is the wave speed.

The functions f and g are determined within constants that sum to zero.

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striking a circular drum

Strike a drum at its center and the vibrations are radially symmetric.

Assuming the drum has radius one, in polar coordinates,

$u(r,t)$ gives deviation from the equilibrium state zero,
at distance r from the center of the drum,
at time t, starting at the strike of the drum.

As the vibrations are radially symmetric,
we have a one dimensional spatial problem.

the wave equation

Let $u(r,t)$ be the deviation from the equilibrium state zero, at distance r from the center of the drum, at time t, starting at the strike of the drum.

We solve the wave equation in polar coordinates

$$\frac{\partial^2 u}{\partial t^2} = \nabla^2 u \quad 0 \leq r < 1, \quad t \geq 0,$$

with initial conditions

1. $u(r,0) = f(r)$, the profile of the drum at $t = 0$, $r < 1$,
2. $u_t(r,0) = 0$, we strike once, at $t = 0$,

and boundary condition

3. $u(1,t) = 0$, for $t \geq 0$, the drum is fixed at end, at $r = 1$.

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separation of variables

In the method of the separation of variables,
we propose $U(r,t)$ as the form of the solution

$$U(r,t) = R(r)T(t),$$

where

- R is a function depending only on r, and
- T is a function depending only on t.

Subjecting U to the conditions of the wave equation $\dfrac{\partial^2 u}{\partial t^2} = \nabla^2 u$
we compute

$$\frac{\partial^2 U}{\partial t^2} = R(r)\frac{\partial^2 T}{\partial t^2} \quad \text{and} \quad \frac{\partial^2 U}{\partial r^2} = \nabla^2 R \cdot T(t).$$

the separation constant λ

The wave equation on

$$\frac{\partial^2 U}{\partial t^2} = R(r) \frac{\partial^2 T}{\partial t^2} \quad \text{and} \quad \nabla^2 U = \nabla^2 R \cdot T(t)$$

gives

$$R(r) \frac{\partial^2 T}{\partial t^2} = \nabla^2 R \cdot T(t) \quad \text{or equivalently} \quad \frac{1}{T(t)} \frac{\partial^2 T}{\partial t^2} = \frac{1}{R(r)} \nabla^2 R.$$

The assumption of $U(r,t) = R(r)T(t)$ for solution implies the existence of a separation constant λ, wedged in between the equation:

$$\frac{1}{T(t)} \frac{\partial^2 T}{\partial t^2} = \lambda = \frac{1}{R(r)} \nabla^2 R.$$

two decoupled ordinary differential equations

By the separation

$$\frac{1}{T(t)}\frac{\partial^2 T}{\partial t^2} = \lambda = \frac{1}{R(r)}\nabla^2 R.$$

we obtain two decoupled ordinary differential equations

$$\frac{\partial^2 T}{\partial t^2} = \lambda T(t),$$

and

$$\nabla^2 R = \lambda R(r).$$

What is $\nabla^2 R$?

the Laplacian in polar coordinates

For $u = u(x, y)$, the Laplacian $\nabla^2 u(x, y) = \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial x^2}$
in polar coordinates, for $u = u(x = r\cos(\theta), y = r\sin(\theta))$, is

$$\nabla^2 u(r, \theta) = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial r^2}{\partial \theta^2}.$$

Do exercise 3 of L-36.

For $u = R(r)$, we then have $\nabla^2 R(r) = \dfrac{\partial^2 R}{\partial r^2} + \dfrac{1}{r}\dfrac{\partial R}{\partial r}$
and we can write

$$\nabla^2 R = \lambda R(r) \quad \text{as} \quad \frac{\partial^2 R}{\partial r^2} + \frac{1}{r}\frac{\partial R}{\partial r} = \lambda R(r).$$

solving for time (*what is λ?*)

As a solution to

$$\frac{\partial^2 T}{\partial t^2} = -\alpha^2 T(t),$$

consider a linear combination of a cosine and a sine function:

$$\begin{array}{rcl} T(t) & = & a\cos(\alpha t) + b\sin(\alpha t) \\ T'(t) & = & -a\alpha\sin(\alpha t) + b\alpha\cos(\alpha t) \\ T''(t) & = & -a\alpha^2\cos(\alpha t) - b\alpha^2\sin(\alpha t) \\ & = & -\alpha^2 T(t). \end{array}$$

Therefore, for some constants a and b,

$$T(t) = a\cos(\alpha t) + b\sin(\alpha t)$$

is a solution to the second order differential equation.

About λ, see Exercise 2 of L-37.

applying the second initial condition

Subject

$$T(t) = a\cos(\alpha t) + b\sin(\alpha t)$$

to the second initial condition

2. $u_t(r, 0) = 0$, we strike once, at $t = 0$,

which gives

$$T'(t) = -a\sin(\alpha t) + b\cos(\alpha t),$$

evaluate at $t = 0$: $T'(0) = b$, thus $b = 0$, and

$$T(t) = a\cos(\alpha t).$$

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solving the wave equation in polar coordinates

The application of separation of variables in polar coordinates to the wave equation is

$$\frac{\partial^2 R}{\partial r^2} + \frac{1}{r}\frac{\partial R}{\partial r} = -\alpha^2 R,$$

or equivalently, after multiplication with r^2:

$$r^2 \frac{\partial^2 R}{\partial r^2} + r\frac{\partial R}{\partial r} + r^2\alpha^2 R = 0.$$

We eliminate α by the substitution $r = z/\alpha$:

$$\left(\frac{z^2}{\alpha^2}\right)\alpha^2\frac{\partial^2 R}{\partial z^2} + \left(\frac{z}{\alpha}\right)\alpha\frac{\partial R}{\partial z} + \left(\frac{z^2}{\alpha^2}\right)\alpha^2 R = z^2\frac{\partial^2 R}{\partial z^2} + z\frac{\partial R}{\partial z} + z^2 R = 0.$$

solving the Bessel equation

The *Bessel equation*

$$z^2 \frac{\partial^2 R}{\partial z^2} + z \frac{\partial R}{\partial z} + z^2 R = 0$$

has two independent solutions:

1. J_0 is the Bessel function of the first kind,
2. Y_0 is the Bessel function of the second kind.

We can verify this using `sympy`.

Bessel functions in Julia are provided
by the package `SpecialFunctions`.

the Bessel function of the first kind

```
julia> using SymPy
julia> z = Sym("z")
julia> J0 = sympy.besselj(0, z)
julia> d1J0 = J0.diff(z)
julia> d2J0 = d1J0.diff(z)

julia> odeJ0 = z^2*d2J0 + z*d1J0 + z^2*J0

julia> sympy.simplify(odeJ0)
0
```

This verifies symbolically that the Bessel functions of the first kind satisfy the ordinary differential equation

$$z^2 \frac{\partial^2 R}{\partial z^2} + z \frac{\partial R}{\partial z} + z^2 R = 0.$$

the Bessel function of the second kind

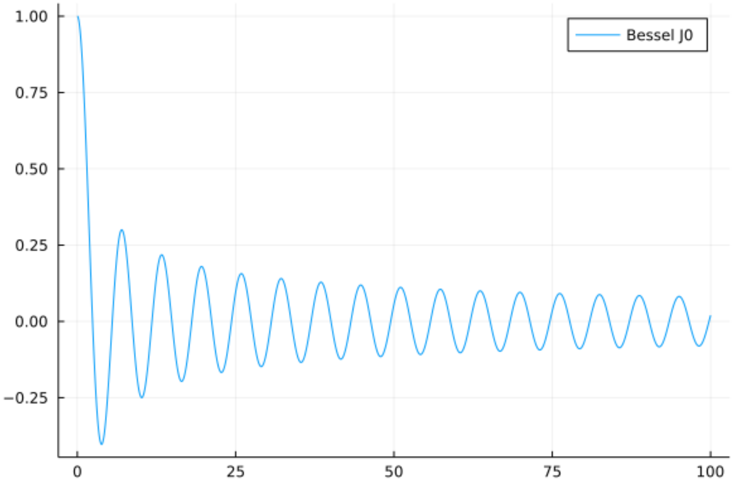
We continue the Julia session:

```
julia> Y0 = sympy.bessely(0, z)
julia> d1Y0 = Y0.diff(z)
julia> d2Y0 = d1Y0.diff(z)

julia> odeY0 = z^2*d2Y0 + z*d1Y0 + z^2*Y0

julia> sympy.simplify(odeY0)
0
```

This verifies symbolically that the Bessel functions of the second kind satisfy the ordinary differential equation

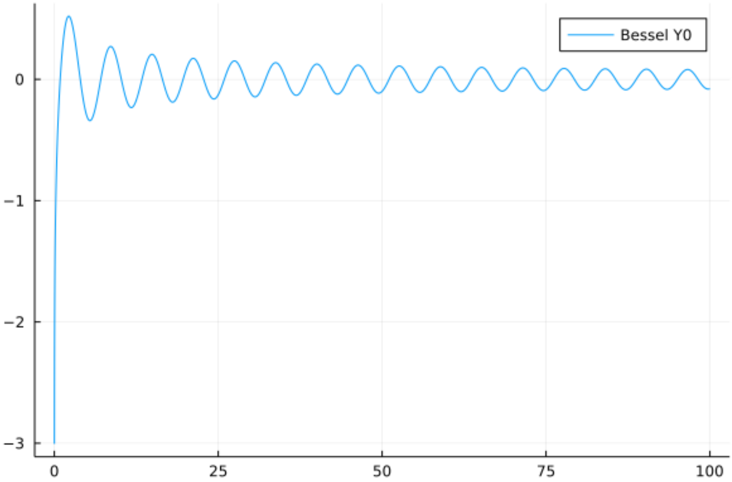
$$z^2 \frac{\partial^2 R}{\partial z^2} + z \frac{\partial R}{\partial z} + z^2 R = 0.$$

the Bessel function of the first kind



We can think of J_0 as a damped version of a cosine.

the Bessel function of the second kind



Observe that Y_0 is no longer bounded near zero.

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imposing the first initial condition

Consider the first initial condition:

1. $u(r, 0) = f(r)$, the profile of the drum at $t = 0$, $r < 1$,

The solution must be bounded,

so we use only J_0, not Y_0.

imposing the boundary condition

Consider the boundary condition:

3. $u(1, t) = 0$, for $t \geq 0$, the drum is fixed at end, at $r = 1$.

The shape of the solution is $U(r, t) = T(t)R(r)$, with

- $T(t) = a\cos(\alpha t)$,
- $R(r) = J_0(z = \alpha r)$,

thus:

$$U(r, t) = a\cos(\alpha t)J_0(\alpha r).$$

The condition $u(1, t) = 0$ gives, setting $r = 1$:

$$\underbrace{a\cos(\alpha t)}_{\neq 0} J_0(\alpha) = 0 \quad \Rightarrow \quad J_0(\alpha) = 0.$$

the first ten roots

```
julia> using FastGaussQuadrature

julia> roots = approx_besselroots(0.0, 10)
10-element Vector{Float64}:
  2.404825557695773
  5.520078110286311
  8.653727912911013
 11.791534439014281
 14.930917708487785
 18.071063967910924
 21.21163662987926
 24.352471530749302
 27.493479132040253
 30.634606468431976

julia>
```

roots of the Bessel functions

Exercise 2:

Consider the Bessel functions J_n, for $n = 0, 1, 2, 3, 4$.

Make a plot of $J_n(z)$, for $z \in [0, 1]$, and for $n = 0, 1, 2, 3, 4$.

What do you observe about the roots of those five functions?

the general solution

For every root α_n of J_0, we have a solution of the form

$$U_n(r, t) = a_n \cos(\alpha_n t) J_0(\alpha_n r).$$

Viewing U_n as a basis of functions, we have the general solution

$$U(r, t) = \sum_{n=1}^{\infty} a_n \cos(\alpha_n t) J_0(\alpha_n r).$$

To determine the coefficients a_n, use the condition

1. $u(r, 0) = f(r)$, the profile of the drum at $t = 0$, $r < 1$.

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an orthogonal basis

Apply $u(r,0) = f(r)$ to $U(r,t) = \displaystyle\sum_{n=1}^{\infty} a_n \cos(\alpha_n t) J_0(\alpha_n r)$ yields

$$U(r,0) = \sum_{n=1}^{\infty} a_n J_0(\alpha_n r) = f(r).$$

With the inner product

$$\langle f(r), g(r) \rangle = \int_0^1 f(r)\, g(r)\, r\, dr.$$

we have

$$\langle J_0(\alpha_n r), J_0(\alpha_m r) \rangle = 0, \quad \text{for } n \neq m.$$

Exercise 3: Verify the orthogonality numerically for $n = 1, 2, 3, 4, 5$.

computing the coefficients

By $\left\langle J_0(\alpha_n r), J_0(\alpha_m r) \right\rangle = 0$, for $n \neq m$,

$$\begin{aligned} \left\langle f(r), J_0(\alpha_m r) \right\rangle &= \left\langle \sum_{n=1}^{\infty} a_n J_0(\alpha_n r), J_0(\alpha_m r) \right\rangle \\ &= a_m \left\langle J_0(\alpha_m r), J_0(\alpha_m r) \right\rangle. \end{aligned}$$

The coefficients a_m are computed as

$$a_m = \frac{\left\langle f(r), J_0(\alpha_m r) \right\rangle}{\left\langle J_0(\alpha_m r), J_0(\alpha_m r) \right\rangle}.$$

a square drum

Exercise 4:
For a square drum, for $x \in [0,1]$ and $y \in [0,1]$,
exercise 11.21 of our text book proposes

$$u(x,y,t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} c_{m,n} \cos\left(\pi t \sqrt{m^2+n^2}\right) \sin(m\pi x) \sin(n\pi y).$$

1. Define the partial differential equation,
   with all initial and boundary conditions for the square drum.
2. Interpret the solution and argue why this proposal makes sense.
3. How are the coefficients $c_{m,n}$ computed?

Can you hear the shape of a drum?

This question is the title of the paper by Mark Kac, published in April 1966 by *The American Mathematical Monthly*, pages 1-23.

Proposal for a Topic of a Project:
This problem is still *open*, according to the paper “Can’t One Really Hear the Shape of a Drum?” by S. Zuluaga and F. Fonseca, in *Acoustical Physics*, Vol. 57, No. 4, pages 465–472, 2011.

- Study the papers and illustrate the importance of this problem for industrial applications.
- Summarize the findings of the two papers.

summary and bibliography

We applied the method of separation of variables to the wave equation in polar coordinates to model the vibrations in a drum.

The circular drum is Problem 5 of Chapter 11 in our text book:

- Charles R. MacCluer: *Industrial Mathematics. Modeling in Industry, Science, and Government*. Prentice Hall, 2000.

The method of separation of variables has limitations.
Numerical methods to solve PDEs are studied in Chapter 8 of the text book in MCS 471:

- Timothy Sauer: *Numerical Analysis*, second edition, Pearson, 2012.